

Name (Print): _____

This exam contains 11 pages (including this cover page) and 9 problems. Check to see if any pages are missing. Print your name above.

No notes will be allowed on the exam. A calculator is permitted on the exam.

Read the following instructions carefully:

- You can only use methods discussed in class so far. i.e., you may **NOT** use any method we haven't learned in class so far.
- **Show ALL your work!** You are being assessed on your work and explanations. Correct answers with little to no work or explanation will not receive full credit. I give partial credits so don't leave a blank if you have some ideas.
- **Organize your work.** Your work should be clearly written and organized. Please use correct mathematical notations when appropriate. For example, please use vector notations when working with vectors and don't use vector notations when working with scalars and points.
- By submitting this exam, you agree that all work represents your individual effort and understanding. Failure to comply with any of the above statements will be considered a violation of the Student Code of Conduct and will result in a 0 for this assessment and possible disciplinary action.

Problem	Points	Score
1	12	
2	12	
3	13	
4	12	
5	13	
6	12	
7	13	
8	13	
9	0	
Total:	100	

Do not write in the table to the right. Adding the bonus, the total score won't exceed 100%.

Please keep in mind that an assessment is just another means of communication between you and me to help me understand what you have learned so far. It is not a measure of your worth as a person nor your intelligence or aptitude as a student. Just give this your best try.

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1. (12 points) Determine the value(s) of h such that the following matrix is the augmented matrix of a **consistent** linear system.

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 2 & 5 & 1 & h \\ 3 & 7 & 0 & 3h - 2 \end{array} \right]$$

2. (12 points) Solve the following system of linear equations using the Gaussian Elimination algorithm:

$$\begin{cases} x + 2y + z = 7 \\ 2x - y + 3z = -2 \\ 3x + y - z = 10 \end{cases}$$

If the system is consistent, write the solution set as a column vector. If the system is inconsistent, explain how you know it.

3. (13 points) Let $A = \begin{bmatrix} 1 & -2 & 1 & 2 \\ 2 & -4 & -1 & 7 \\ -1 & 2 & 2 & -5 \\ 3 & -6 & 1 & 8 \end{bmatrix}$. Describe all solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$ in **parametric form**.

4. (12 points) Is the following set of vectors linearly independent?

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

If they are not linearly independent, find the *smallest* subset that spans the same space.

5. (13 points) Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be a linear transformation defined by

$$T(\mathbf{x}) = A\mathbf{x}, \quad \text{where } A = \begin{bmatrix} 1 & -4 & 7 & -5 \\ 0 & 1 & -4 & 3 \\ 2 & -6 & 6 & -4 \end{bmatrix}$$

- (a) Find all $\mathbf{x} \in \mathbb{R}^4$ such that $A\mathbf{x} = \mathbf{0}$.

- (b) Is $\mathbf{b} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ an image of T ? Justify your reasoning.

6. (12 points) Let $A = \begin{bmatrix} 2 & 1 \\ 0 & -3 \\ 1 & 0 \\ -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$. Multiply the matrices as indicated if possible. If not possible, explain why.

(a) AB

(b) BA

7. (13 points) Let $A = \begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$.

(a) Find A^{-1} , the inverse of the matrix A .

(b) Solve the matrix equation $A\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ using A^{-1} you found in part (a).

8. (13 points) Let $A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ -2 & -3 & 1 & 4 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & -1 & 3 \end{bmatrix}$ and $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be the linear transformation

defined by $T(\mathbf{x}) = A\mathbf{x}$. Answer the following questions and justify your reasoning.

- (a) Do the columns of A span $\mathbb{R}^4$?
- (b) Are the columns of A linearly independent?
- (c) Is T a surjection?
- (d) Is T an injection?

9. (10 points (bonus)) **This is a BONUS problem.**

Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a linear transformation such that

$$T\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} \quad \text{and} \quad T\left(\begin{bmatrix} 7 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} -3 \\ 5 \\ 1 \end{bmatrix}$$

Find the standard matrix A that defines this linear transformation. That is, find A such that $T(\mathbf{x}) = A\mathbf{x}$.

Hint: Recall we can find the standard matrix if we know what T does to the standard basis vectors. What are $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$ using the linearity property?